A PROJECT REPORT ON

AI-POWERED SECURE CHAT SYSTEM WITH AUTOMATED STEGANOGRAPHY DETECTION

Submitted in partial fulfillment of the requirements for the award of the degree

Of

BACHELOR OF TECHNOLOGY

In

CSE (ARTIFICIAL INTELLIGENCE)

Under the Esteemed Guidance of

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SREENIVASA INSTITUTE OF TECHNOLOGY AND MANAGEMENT STUDIES

(Autonomous)

(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu),

Chittoor, A.P-517127.

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE)

(2025 -26)

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CERTIFICATE

This is to certify that the project work entitled **"AI-POWERED SECURE CHAT SYSTEM WITH AUTOMATED STEGANOGRAPHY DETECTION"** is a bonafide work of

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Submitted for University Examination (Viva-Voice) held on……

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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Institute Mission

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M2: Ignite the students to acquire self-reliance in the latest technologies.

M3: Unleash and encourage the innate potential and creativity of students.

M4: Inculcate confidence to face and experience new challenges.

M5: Foster enterprising spirit among students.

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- Ignite the students to acquire self-reliance in the latest technologies.
- Foster Enterprising spirit among students.

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PROGRAMME EDUCATIONAL OBJECTIVES(PEO's):

PEO1: Expertise with computer science and Engineering, artificial intelligence and machine learning disciplines through quality studies, enabling success in IT industries. **(Professional Competency)**

PEO2: Establish start-up companies or employed in reputed computing industries or government sectors or pursue higher studies in the domain of CSE (AI) **(Successful Career Goals)**

PEO3: Enhance knowledge by updating advanced technological concepts for facing the rapidly changing world and contribute to society through innovation and creativity **(Continuing Education and Contribution to Society)**.

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PSO2: Deploy the artificial intelligence techniques to perform human intelligence in creating innovative products and solutions.

PROGRAMME OUTCOMES (PO's):

PO1-Engineeringknowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 - Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 - Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 - Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 - Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 - The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 – Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 -Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 - Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10-Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 - Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 - Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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Course Out Comes for Project Work

On completion of project work the student will be able to

CO1. Demonstrate in-depth knowledge on the project topic.

CO2. Identify, analyze and formulate complex problem chosen for project work to attain substantiated conclusions.

CO3. Design solutions to the chosen project problem.

CO4. Undertake investigation of project problem to provide valid conclusions.

CO5. Use the appropriate techniques, resources and modern engineering tools necessary for project work.

CO6. Apply project results for sustainable development of the society.

CO7. Understand the impact of project results in the context of environmental sustainability.

CO8. Understand professional and ethical responsibilities while executing the project work.

CO9. Function effectively as individual and a member in the project team.

CO10. Develop communication skills, both oral and written for preparing and presenting project report.

CO11. Demonstrate knowledge and understanding of cost and time analysis required for carrying out the project.

CO12. Engage in lifelong learning to improve knowledge and competence in the chosen area of the project.

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CO – PO MAPPING

CO\PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO.1	3	-	-	-	-	-	-	-	-	-	-	-	3	3
CO.2	-	3	-	-	-	-	-	-	-	-	-	-	3	3
CO.3	-	-	3	-	-	-	-	-	-	-	-	-	3	3
CO.4	-	-	-	3	-	-	-	-	-	-	-	-	3	3
CO.5	-	-	-	-	3	-	-	-	-	-	-	-	3	3
CO.6	-	-	-	-	-	3	-	-	-	-	-	-	3	3
CO.7	-	-	-	-	-	-	3	-	-	-	-	-	3	3
CO.8	-	-	-	-	-	-	-	3	-	-	-	-	3	3
CO.9	-	-	-	-	-	-	-	-	3	-	-	-	3	3
CO.10	-	-	-	-	-	-	-	-	-	3	-	-	3	3
CO.11	-	-	-	-	-	-	-	-	-	-	3	-	3	3
CO.12	-	-	-	-	-	-	-	-	-	-	-	3	3	3
CO	3	3	3	3	3	3	3	3	3	3	3	3	3	3

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Evaluation Rubrics for Project Work

Rubric(CO)	Excellent(wt=3)	Good(wt=2)	Fair(wt=1)
Selection of Topic (CO1)	Selected a latest topic through complete knowledge of facts and Concepts	Selected a topic through partial knowledge off acts and concepts	Selected a topic through improper knowledge of facts and concepts
Analysis and Synthesis (CO2)	Thorough Comprehension through analysis/ synthesis	Reasonable Comprehension through analysis/ synthesis	Improper Comprehension through analysis/ synthesis
Problem Solving (CO3)	Thorough comprehension about What is proposed in the literature papers	Reasonable comprehension about What is proposed in the literature papers	Improper comprehension about What is proposed in the literature
Literature Survey (CO4)	Extensive literature Survey with standard References	Considerable literature Survey with standard References	Incomplete literature Survey with substandard References
Usage of Techniques & Tools (CO5)	Clearly identified and has complete knowledge Of techniques & tools used in the project work	Identified and has sufficient knowledge of techniques & tools used in the project work	Identified and has in adequate knowledge of techniques & tools used in project work
Project work impact on Society (CO6)	Conclusion of project work has strong impact on society	Conclusion of project work has considerable impact on society	Conclusion of project work has feeble impact on society
Project work impact on Environment (CO7)	Conclusion of project work has strong impact on Environment	Conclusion of project work has considerable impact on environment	Conclusion of project work has feeble impact on environment
Ethical attitude (CO8)	Clearly understands ethical and social practices.	Moderate understanding of ethical and social practices.	Insufficient understanding of ethical and social practices.
Independent earning (CO9)	Did literature survey and selected topic with little Guidance	Did literature survey and selected topic with considerable guidance	Selected a topics suggested by the Supervisor
Oral Presentation (CO10)	Presentation in logical sequence with key points, clear conclusion and excellent language	Presentation with key points, conclusion and good language	Presentation with insufficient key points and improper Conclusion
Report Writing (CO10)	Status report with clear and logical sequence of Chapters using excellent language	Status report with logical sequence of chapters using understandable language	Status report not properly organized
Time and Cost Analysis (CO11)	Comprehensive time and cost analysis	Moderate time and cost analysis	Reasonable time and cost analysis
Continuous learning (CO12)	Highly enthusiastic Towards continuous Learning	Interested in continuous learning	Inadequate interest in continuous learning

ACKNOWLEDGEMENT

A Project of this magnitude would have not been possible without the guidance and co- ordination of many people. I am fortune in having top quality people to help, support and guide us in every step towards our goal.

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We express our sincere thanks to our guide **Ms. M. Deepika.,** for offering us the opportunity to do this work under his guidance.

We express our sincere salutation to all other teaching and non-teaching staff of our department for their direct and indirect support given during our project work. Last but not the least, we dedicate this work to our parents and the Almighty who have been with us throughout and helped us to overcome the hard times.

DECLARATION

We certify that

- The work contained in this report is original and has been done by me under the Guidance of my supervisor.
- The work has not been submitted to any other Institute for any degree or diploma.
- We have followed the guidelines provided by the Institute in preparing the report.
- We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- Whenever We have used materials (data, theoretical analysis, figures, and text) from other sources, We have given due credit to them by citing them in the text of the report and giving their details in the references. Further, We have taken permission from the copyright owners of the sources, whenever necessary.

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ABSTRACT

With the rapid growth of digital communication, images have become a common medium for covert data transmission using steganography techniques. Among these, Least Significant Bit (LSB) steganography is widely used due to its simplicity and high embedding capacity, making it a serious cybersecurity concern. This project presents a secure chat application that enables the exchange of text messages and images while automatically detecting hidden data within transmitted images. Every received image is analyzed at the server using steganographic detection techniques to identify concealed payloads. If hidden data is detected, the system extracts the payload and examines whether it contains programming code or malicious content. The proposed system avoids aggressive image resizing to preserve subtle pixel-level artifacts, thereby improving detection accuracy. Upon identifying suspicious content, the system alerts the user and blocks the image delivery. This approach enhances real-time communication security and prevents covert malware transmission through digital images.

Keywords:

Steganography, LSB, Secure Messaging, Image Analysis, Hidden Data Detection, Cyber Security

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LIST OF ABBREVATIONS

ABBREVIATIONS	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DB	Database
DL	Deep Learning
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
JSON	JavaScript Object Notation
LSB	Least Significant Bit
ML	Machine Learning
REST	Representational State Transfer
SRNet	Spatial Rich Model Network
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator
UUID	Universally Unique Identifier

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